Participation E 1

For special participation part E, I asked DeepSeek to teach me how to write a smart contract on BTC chain: https://chat.deepseek.com/share/9wu5248zp1z8hls9xy

Based on our conversation, here is a summary in English of what I have accomplished and the current situation:

Progress Summary

I have successfully completed the initial setup and are very close to deploying my first Bitcoin smart contract. Here is what I have done:

1. ✅ **Environment Setup**: I installed all prerequisite tools (`Node.js` , `NPM` , `Git`) on my Windows 11 computer.
2. ✅ **Core Tools Installation**: I installed the global CLI tool (`@opcat-labs/cli-opcat`) which is used to create and manage projects.
3. ✅ **Project Creation**: I created a new project named `helloworld` using the CLI. This command automatically generated the complete project structure and configuration files.
4. ✅ **Project Dependencies (SDK)**: I entered the project directory and ran `npm install` . This installed all necessary dependencies for the project, including the crucial `@opcat-labs/scrypt-ts-opcat` **SDK**—the library containing the functions and classes needed to write my contract.
5. ✅ **Smart Contract Code**: I correctly modified the main contract file (`src/contracts/helloworld.ts`), creating a simple contract that stores a hash and can only be unlocked by providing the matching message.
6. ✅ **Contract Compilation**: I ran `npm run compile` , which successfully transformed my TypeScript code into a blockchain-readable format and generated a contract artifact file (`artifacts/contracts/helloworld.json`).
7. ✅ **Test Key Generation**: I generated a private key for the Bitcoin testnet using `npm run genprivkey` . The command provided I with a testnet address (`mx79K19aq2Mq9mjUL1SHQ4y5NwTwcLKsFG`) to receive faucet funds.

Current Blocking Issue

I am currently stuck at the final deployment step. When I run the deployment script (`npx tsx deploy.ts`), it fails with the following error:

```
Error: Cannot find module 'D:\...\helloworld\dist\cjs\index.cjs'
```

Analysis of the Problem:
This is a technical configuration or compatibility issue, not a mistake in my following of the tutorial steps. The most likely causes are:

- **Node.js Version Mismatch**: I am using Node.js v22.20.0. The tools in this project (`tsx` and the SDK) may not be fully compatible with this very recent version and expect an older Long-Term Support (LTS) version like Node.js 18 or 20.
- **Incomplete Build Process**: The error points to a missing file in a `dist` directory. my `npm run compile` command compiled the *sCrypt contract* itself, but the deployment script may require the *TypeScript project code* to be compiled into a `dist` folder, which hasn't happened yet.

Immediate Next Steps & Solution

To resolve this and successfully deploy my contract, I need to:

1. **Add `loadArtifact()` to my Script**: First, ensure my `deploy.ts` file includes the two critical lines that tell the system how to interact with my compiled contract. Add them after the other import statements:

   ```
   import artifact from './artifacts/contracts/helloworld.json'
   Helloworld.loadArtifact(artifact)
   ```

2. **Switch to a Compatible Node.js Version (Recommended Solution)**: The most effective fix is to switch from Node.js v22 to an LTS version (v18 or v20). I can do this using a version manager like `nvm-windows` for Windows.
3. **Reinstall Dependencies & Recompile**: After switching Node.js versions, go back to my project folder, delete the `node_modules` folder, and run `npm install` followed by `npm run compile` again to ensure everything is built correctly for the new environment.

Once these steps are complete, funding my test address from the faucet and running `npx tsx deploy.ts` should successfully deploy and call my "Helloworld" smart contract on the Bitcoin testnet.

In short: I have perfectly followed all tutorial steps. The final error is a common technical hurdle that is resolved by adjusting my Node.js version and verifying my deployment script's code.